

# Question 2 Clustering: Full Trait Model

This notebook evaluates the custom CSC 466 random forest using `data/processed/breed_clustering_full.csv`.

This notebook clusters dog breeds using the full set of numeric AKC trait scores.

## Cluster Summary on AKC Groups

Overall Metrics  
Overall silhouette score: 0.1  
Rand Index: 0.76  
Average radius / average intercluster distance: 1.33

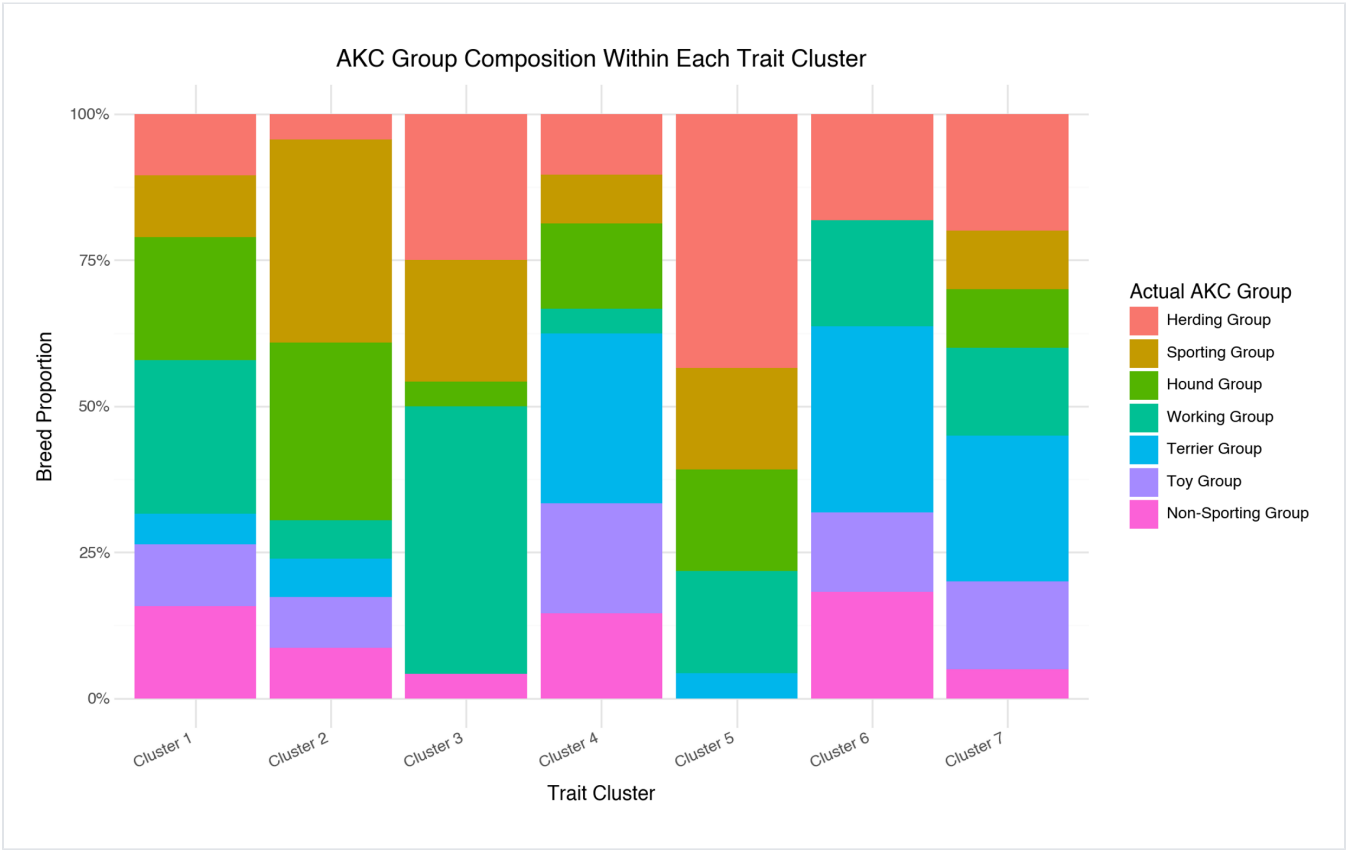
| Trait Cluster                   | Cluster 1     | Cluster 2      | Cluster 3     | Cluster 4     | Cluster 5     |
|---------------------------------|---------------|----------------|---------------|---------------|---------------|
| Metric                          |               |                |               |               |               |
| Cluster Size                    | 19            | 46             | 24            | 48            | 13            |
| Most Common AKC Group           | Working Group | Sporting Group | Working Group | Terrier Group | Herding Group |
| Most Common AKC Group Count     | 5             | 16             | 11            | 14            | 3             |
| Predicted AKC Group             | Working Group | Sporting Group | Working Group | Terrier Group | Herding Group |
| Distance To Predicted AKC Group | 2.14          | 1.35           | 1.14          | 0.96          | 1.33          |
| AKC Rand Index                  | 0.67          | 0.64           | 0.72          | 0.62          | 0.65          |
| Silhouette Score                | 0.1           | 0.08           | 0.06          | 0.11          | 0.07          |
| Radius                          | 3.47          | 3.7            | 4.34          | 4.51          | 3.5           |

## Cluster Centroids vs AKC Group Traits Averages

|                            | Cluster 1 | Cluster 2 | Cluster 3 | Cluster 4 | Cluster 5 | Cluster 6 | Cluster 7 |
|----------------------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| Trait                      |           |           |           |           |           |           |           |
| Affectionate With Family   | 4.11      | 4.78      | 4.96      | 4.73      | 3.00      | 4.86      | 4.55      |
| Good With Young Children   | 3.00      | 4.61      | 4.12      | 3.31      | 3.17      | 4.91      | 3.70      |
| Good With Other Dogs       | 3.26      | 4.67      | 3.25      | 3.04      | 3.17      | 3.59      | 3.15      |
| Shedding Level             | 2.63      | 2.85      | 3.12      | 2.48      | 2.83      | 2.59      | 1.30      |
| Coat Grooming Frequency    | 2.05      | 1.98      | 1.71      | 1.92      | 2.26      | 3.23      | 3.60      |
| Drooling Level             | 2.42      | 1.74      | 2.92      | 1.40      | 1.26      | 1.36      | 1.80      |
| Openness To Strangers      | 2.84      | 4.00      | 3.38      | 3.46      | 2.74      | 4.41      | 2.95      |
| Playfulness Level          | 3.05      | 4.02      | 3.67      | 3.56      | 3.09      | 4.32      | 3.50      |
| Watchdog/Protective Nature | 3.95      | 3.00      | 4.42      | 3.77      | 3.70      | 3.91      | 4.00      |
| Adaptability Level         | 3.26      | 4.07      | 3.92      | 3.75      | 3.09      | 4.36      | 3.70      |
| Trainability Level         | 3.16      | 4.00      | 4.29      | 3.69      | 4.04      | 4.18      | 3.55      |
| Energy Level               | 3.11      | 4.09      | 4.08      | 3.52      | 4.04      | 3.86      | 3.25      |
| Barking Level              | 1.63      | 3.09      | 2.71      | 3.71      | 2.87      | 4.14      | 3.10      |
| Mental Stimulation Needs   | 3.16      | 3.85      | 4.17      | 3.42      | 3.52      | 3.95      | 3.65      |

|                            | AKC Group | Herding Group | Hound Group | Non-Sporting Group | Sporting Group | Terrier Group | Toy Group | Working Group |
|----------------------------|-----------|---------------|-------------|--------------------|----------------|---------------|-----------|---------------|
| Trait                      |           |               |             |                    |                |               |           |               |
| Affectionate With Family   | 4.18      | 4.25          |             | 4.85               | 4.64           | 4.68          | 4.76      | 4.44          |
| Good With Young Children   | 3.70      | 3.78          |             | 4.00               | 4.12           | 3.97          | 3.57      | 3.91          |
| Good With Other Dogs       | 3.36      | 4.22          |             | 3.40               | 3.91           | 2.94          | 3.76      | 3.22          |
| Shedding Level             | 2.82      | 2.50          |             | 2.50               | 2.70           | 2.16          | 2.33      | 2.97          |
| Coat Grooming Frequency    | 2.42      | 1.66          |             | 2.50               | 2.33           | 2.35          | 2.57      | 2.22          |
| Drooling Level             | 1.55      | 1.75          |             | 1.65               | 2.12           | 1.29          | 1.10      | 2.66          |
| Openness To Strangers      | 3.12      | 3.22          |             | 3.80               | 3.73           | 3.65          | 3.81      | 3.31          |
| Playfulness Level          | 3.64      | 3.38          |             | 3.80               | 3.85           | 3.68          | 3.81      | 3.53          |
| Watchdog/Protective Nature | 3.91      | 3.06          |             | 3.80               | 3.18           | 3.94          | 3.62      | 4.53          |
| Adaptability Level         | 3.70      | 3.53          |             | 3.90               | 3.94           | 3.81          | 4.14      | 3.62          |
| Trainability Level         | 4.24      | 3.25          |             | 3.60               | 4.64           | 3.52          | 3.67      | 3.91          |
| Energy Level               | 4.06      | 3.72          |             | 3.50               | 4.09           | 3.61          | 3.29      | 3.69          |
| Barking Level              | 3.39      | 3.31          |             | 2.90               | 2.73           | 3.55          | 3.33      | 2.78          |
| Mental Stimulation Needs   | 4.00      | 3.62          |             | 3.40               | 3.82           | 3.61          | 3.29      | 3.72          |

## How well do trait-based clusters line up with AKC Groups?



## Prediction Output

This table saves each breed's assigned cluster with its average rank, popularity tier, and AKC group.

|     | Breed                       | Trait Cluster | Predicted AKC Group | Actual AKC Group   | Distance To Predicted AKC Group | Correct AKC Prediction |
|-----|-----------------------------|---------------|---------------------|--------------------|---------------------------------|------------------------|
| 0   | Affenpinscher               | Cluster 4     | Terrier Group       | Toy Group          | 0.96                            | False                  |
| 1   | Afghan Hound                | Cluster 7     | Terrier Group       | Hound Group        | 1.88                            | False                  |
| 2   | Airedale Terrier            | Cluster 7     | Terrier Group       | Terrier Group      | 1.88                            | True                   |
| 3   | Akita                       | Cluster 5     | Herding Group       | Working Group      | 1.80                            | False                  |
| 4   | Alaskan Malamute            | Cluster 5     | Herding Group       | Working Group      | 1.80                            | False                  |
| ... | ...                         | ...           | ...                 | ...                | ...                             | ...                    |
| 197 | Wire Fox Terrier            | Cluster 6     | Terrier Group       | Terrier Group      | 2.14                            | True                   |
| 198 | Wirehaired Pointing Griffon | Cluster 3     | Working Group       | Sporting Group     | 1.14                            | False                  |
| 199 | Wirehaired Vizsla           | Cluster 2     | Sporting Group      | Sporting Group     | 1.35                            | True                   |
| 200 | Xoloitzcuintli              | Cluster 4     | Terrier Group       | Non-Sporting Group | 0.96                            | False                  |
| 201 | Yorkshire Terrier           | Cluster 6     | Terrier Group       | Toy Group          | 2.14                            | False                  |

202 rows × 6 columns